



Optimal Delivery Route

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Difficulty Level : Easy Medium

Description

Ali works as a delivery driver for a logistics company. He needs to deliver packages to different cities connected by roads with known travel times. Your task is to help Ali find the minimum time needed to reach every city starting from the warehouse.

Problem Statement

You are given a map of N cities and M bidirectional roads connecting them. Each road has a travel time (positive integer).

Ali starts from city S (the warehouse). You need to compute the **minimum travel time** from city S to **every other city**

If a city is unreachable, return -1 for that city.

Input Format

- An integer N — the number of cities ($1 \leq N \leq 1000$)
- An integer M — the number of roads
- M lines each containing three integers u, v, w — indicating a road between city u and city v with travel time w ($1 \leq u, v \leq N, 1 \leq w \leq 10^4$)
- An integer S — the starting city

Output Format

Print a single line with N space-separated integers: the shortest time to reach each city from S.

If a city is unreachable, print -1 for that city. The travel time to city S itself should be 0.

Constraints

- $1 \leq N \leq 1000$
- $0 \leq M \leq N * (N - 1) / 2$
- No multiple edges between the same pair of cities.

Examples

Example 1

Input	Output
5 6 1 2 2 1 3 4 2 3 1 2 4 7 3 5 3 4 5 1 1	0 2 3 9 6

Example 2

Input	Output
4	0

2 1 2 5 3 4 2	5 -1 -1
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Explanation: Cities 3 and 4 are not reachable from city 1.

Example 3

Input	Output
1 0 1	0

Explanation: Only one city exists and it's the starting city.